Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. We trained each model for 100 epochs with a batch size of 128.

2 Results

Figure 1 shows the test accuracy of ResNet-50 by dataset.

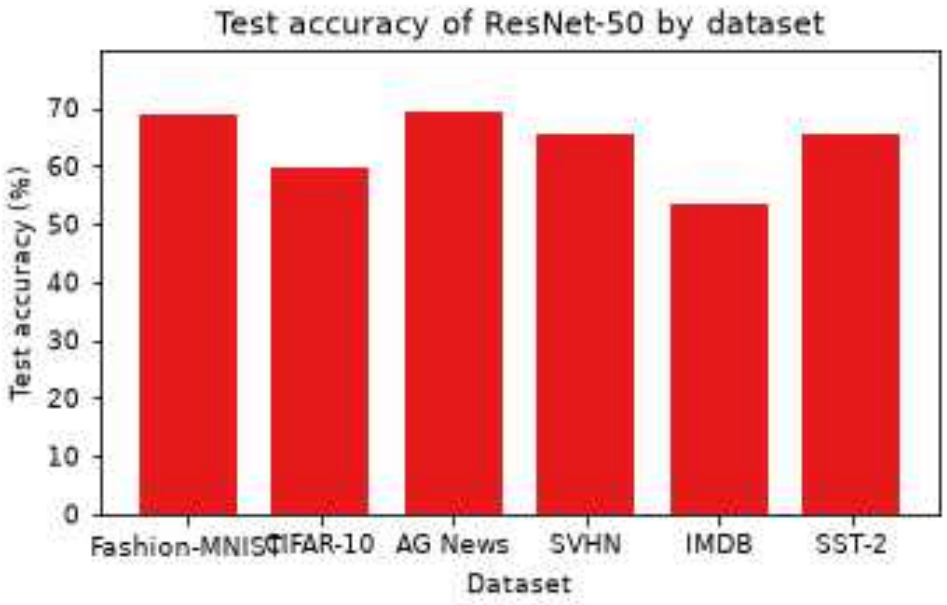


Figure 1: Test accuracy of ResNet-50 by dataset for the models in this experiment.

ResNet-50 reaches over 50% test accuracy on IMDB. ResNet-50 reaches 81.4% test accuracy on SVHN. ResNet-50 scores 27.9 points higher on SVHN than on IMDB (Figure 1).

Figure 2 shows the validation accuracy of CNN by dataset.

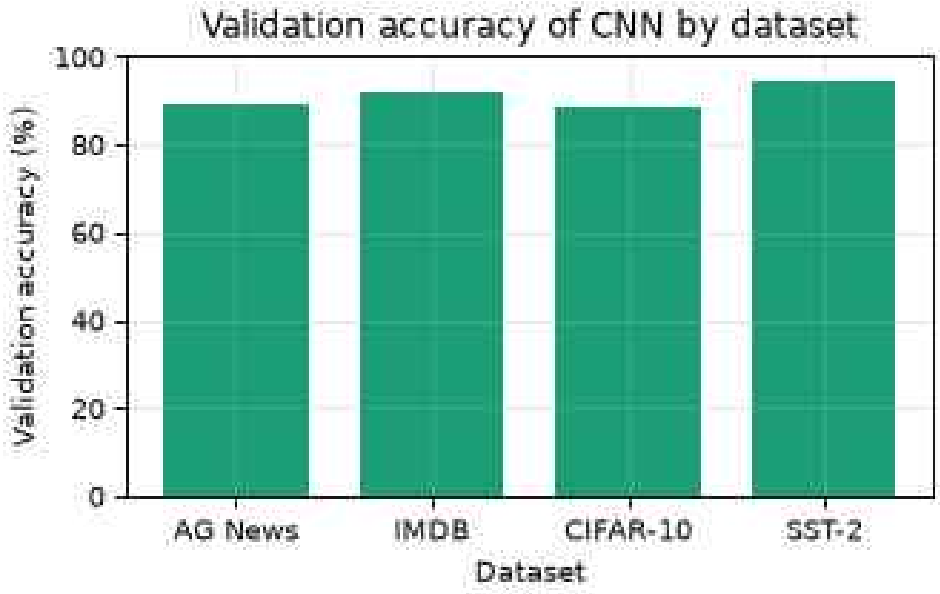


Figure 2: Validation accuracy of CNN by dataset for the models in this experiment.

CNN achieves 44.6% validation accuracy on SST-2. CNN scores 55.6 points higher on SST-2 than on AG News.

Figure 3 shows the accuracy by dataset.

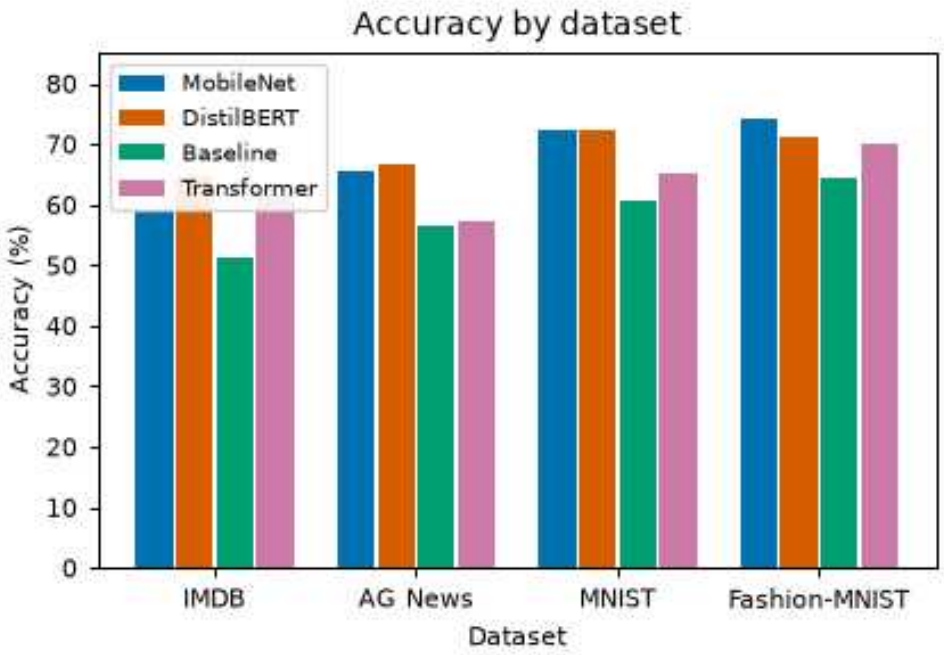


Figure 3: Accuracy by dataset for the models in this experiment.

As shown in Figure 3, Transformer reaches 57% accuracy on AG News. Baseline reaches 57.3% accuracy on AG News (Figure 3).

3 Discussion

Random Forest reaches 72.9% accuracy on MNIST. We note that the training was stable for all models.